AI-Powered Face Recognition & Attendance System

Interactive Technical Project Report & Documentation

Project Name:	Face Recognition & Attendance System	**Project Directory:**	`C:\Advancejava\FaceRecognitionSystem`
Technology Stack:	Java 25, Spring Boot 3.2.5, OpenCV 4.9, MySQL 8.0	**Live App Link:**	http://localhost:8080
Report Date:	September 26, 2026	**Status:**	**ACTIVE & RUNNING**

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1. Executive Summary

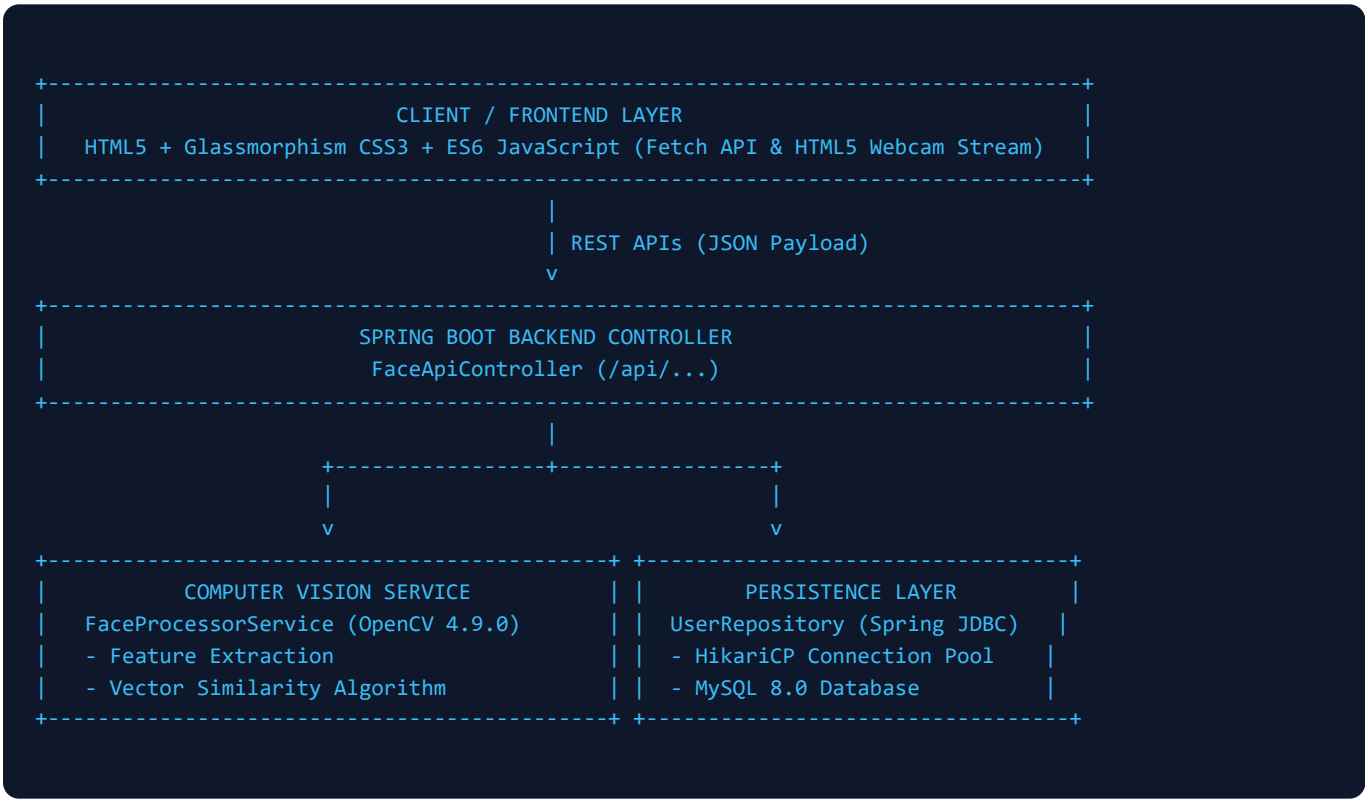
Project Overview: The AI Face Recognition & Attendance System is an enterprise web application engineered for contactless identity recognition, automated attendance logging, and workforce management. Built using Java 25, Spring Boot 3, and OpenCV Computer Vision, the platform extracts facial feature vectors from video frames, executes similarity matching against stored MySQL templates, and provides an interactive Admin Control Portal for workforce metrics and account management.

Direct Link to Live Application: http://localhost:8080

2. Key Features & Functionalities

- **Real-Time Facial Identification:** Captures live video streams via HTML5 webcam, extracts facial feature vectors using OpenCV OpenPNP (OpenPNP OpenCV 4.9.0), and calculates similarity matching with a 70%+ threshold.
- **Automated Attendance Logging:** Automatically generates timestamped records in the MySQL `attendance_logs` table upon successful match verification.
- **User Enrollment & Face Training:** Snapshot capture interface allowing new users to enroll with full name, email address, company/department, and instant facial embedding encoding.
- **Admin Control Center & Workforce Analytics:**
  - Total enrolled users & company department statistics summary cards.
  - Visual grid displaying user distribution metrics per department/company.
  - Department filter dropdown for user directory inspection.
  - **Update User:** Modal dialog to edit name, email, and department (**`PUT /api/users/{id}`**).
  - **Delete User:** Instant deletion of user records from MySQL with foreign key cascading (**`DELETE /api/users/{id}`**).

3. System Architecture



4. Technology Stack Matrix

Layer	Technology	Version	Documentation Link
Runtime	Java JDK	25.0.2 / 17+	Oracle Java Documentation
Framework	Spring Boot	3.2.5	Spring Boot Docs

Layer	Technology	Version	Documentation Link
Computer Vision	OpenCV OpenPNP	4.9.0-0	OpenCV Official Site
Database	MySQL Server	8.0+	MySQL Developer Docs
Data Access	Spring JDBC Template	3.2.5	Spring JDBC Docs
Frontend UI	HTML5, Glassmorphism CSS3, JS	ES6+	Open App UI

5. Interactive REST API Specifications

Method	Endpoint	Request Payload	Description
GET	`/api/health`	None	System health status & enrolled user count.
POST	`/api/register`	`{ name, email, department, imageBase64 }`	Enrolls user, extracts face features, saves to MySQL.
POST	`/api/recognize`	`{ imageBase64 }`	Analyzes webcam frame & logs attendance.
GET	`/api/users`	None	Returns list of all enrolled users.
GET	`/api/logs`	None	Retrieves recent attendance recognition logs.
GET	`/api/admin/stats`	None	Returns company metrics & department user counts.
PUT	`/api/users/{id}`	`{ name, email, department }`	Updates user details in MySQL.
DELETE	`/api/users/{id}`	None	Deletes user account and associated attendance records.

6. Database Schema Design

Table 1: `users`

Field Name	Data Type	Constraints	Description
`id`	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Unique User ID
`name`	VARCHAR(100)	NOT NULL	Employee Name
`email`	VARCHAR(100)	NOT NULL, UNIQUE	Email Address
`department`	VARCHAR(100)	DEFAULT 'General'	Company / Department
`face_features`	LONGTEXT	NULLABLE	JSON Array of Extracted Features
`photo_base64`	LONGTEXT	NULLABLE	Base64 Photo Snapshot
`created_at`	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Registration Timestamp

Table 2: `attendance_logs`

Field Name	Data Type	Constraints	Description
`id`	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Log Entry ID
`user_id`	BIGINT	FOREIGN KEY (users.id)	User ID Reference
`user_name`	VARCHAR(100)	NOT NULL	Matched Name
`department`	VARCHAR(100)	NULLABLE	Company Department
`confidence_score`	DOUBLE	NOT NULL	Match Confidence %
`status`	VARCHAR(50)	DEFAULT 'SUCCESS'	Status Tag
`timestamp`	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Access Timestamp

7. Execution & Access Guide

To access the running application:

- Click to open in browser: **http://localhost:8080**
- Admin Stats Endpoint: **http://localhost:8080/api/admin/stats**
- Users List Endpoint: **http://localhost:8080/api/users**
